The Qungasvik-Healthy Hearts Intervention: A culturally grounded and evidence-based intervention to support increased physical activity, a traditional Yup'ik diet, and social connection for Elders and youth in Alaska

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Introduction

The remoteness and lack of acute care in rural Alaska Native (AN) communities, means that cardiovascular and cerebrovascular events are more likely to be catastrophic and have lower survivorship and recovery outcomes. Due to the remoteness, transportation challenges, and lack of access to medicine and healthcare, preventative health for chronic disease in rural AN communities is of utmost concern.

During Phase I of the American Indian, Alaska Native, Native Hawaiian, Pacific Islander (AIAN NHPI) Enrichment Initiative, our Alaska site conducted a pilot implementation of the adapted culturally-grounded and evidence-based intervention known as **Qungasvik-Healthy Hearts** to assess feasibility and acceptability, as well as preliminary impact on cerebrovascular and cardiovascular disease risk in one rural AN community.

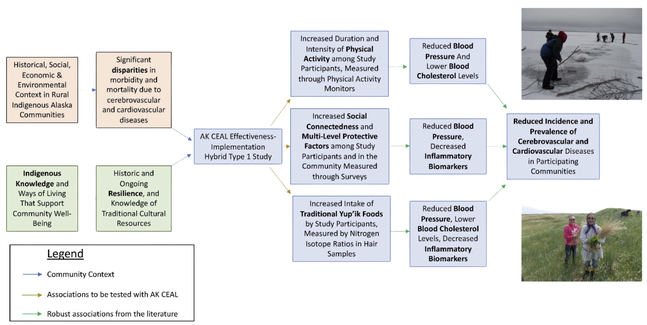


Figure 1. Conceptual Framework for Phase 1 of Qungasvik - Healthy Hearts

Objectives

- **Engage** with collaborating AN communities and establish the AK EI Regional Working Group (RWG).
- **Adapt** an existing evidence-based intervention to expand reach to young people and older adults.
- **Measure** impacts on upstream risk factors for the prevention of cerebrovascular and cardiovascular diseases.
- **Assess** community-level contextual factors impacting intervention adoption to inform the trial implementation.

Methodology

- This study piloted the adapted Qungasvik - Healthy Hearts intervention in one rural AN community with young people and older adults in an effectiveness-implementation hybrid type 1 study.
- The data in figure 2 was collected at three different timepoints during the pilot (baseline, midpoint, and endpoint): surveys, blood pressure, traditional food intake (hair samples), physical activity (actigraphs).
- Intervention activities were submitted by coordinators throughout the duration of the pilot.

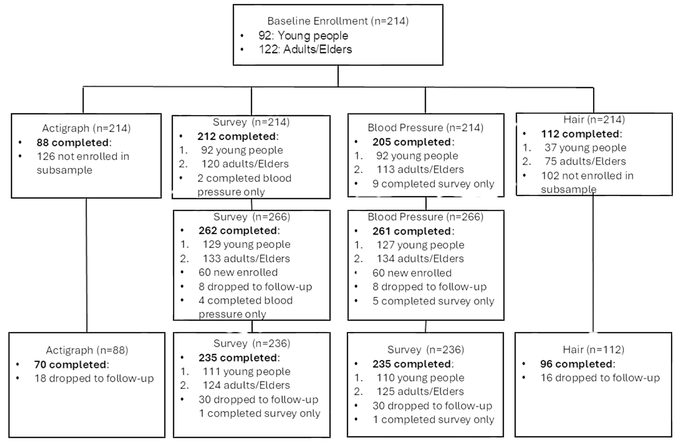


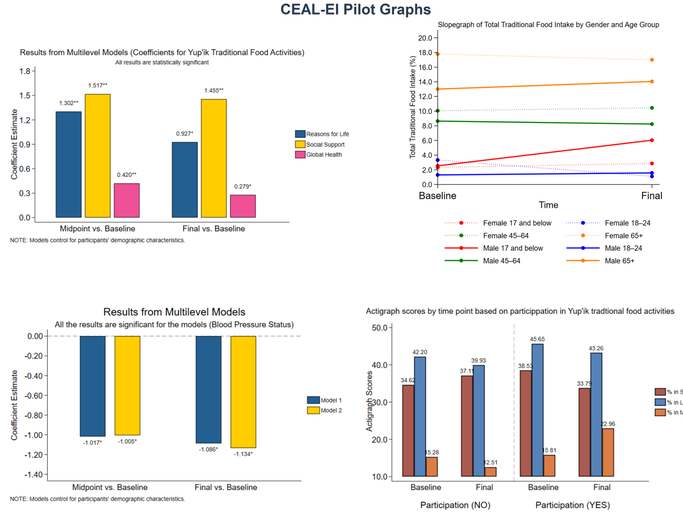
Figure 2. The CONSORT diagram shows results from the completed pilot.



Figure 3. Qungasvik intervention Modules

Results

The AK EI Phase I pilot study show that higher participation in the Qungasvik-Healthy Hearts intervention is associated with improved reasons for life, social support and global health and decreased blood pressure. Results also show promising impacts of the Qungasvik-Healthy Hearts intervention on physical activity and traditional food intake. Activities that contribute most highly to these improvements include participation in Yup'ik traditional food activities and engagement with the Community Planning Group (CPG). Overall, research activities on Phase I accomplished our primary specific aims to pilot the Qungasvik-Healthy Hearts adapted intervention, and to demonstrate that our approach was acceptable and feasible within the context of rural Alaska Native communities.



Conclusion

- The need for a culturally appropriate intervention designed to mitigate cerebrovascular and cardiovascular risk factors is urgent. At both baseline and midpoint, approximately 70% of adults who participated in the study had hypertensive blood pressure readings, indicating an alarming need for sustainable and effective intervention.

- The pilot demonstrated that the Qungasvik-Healthy Hearts intervention is both feasible and highly acceptable. Community members showed strong enthusiasm for participation and for data collection procedures, which required protocol refinements but were logistically feasible.

Key Sources & Acknowledgements

Rasmus, S. M., Charles, B., & Mohatt, G. V. (2014). Creating Qungasvik (a Yup'ik intervention "toolbox"): Case examples from a community-developed and culturally-driven intervention. American Journal of Community Psychology, 54(1-2), 140–152. List the primary sources used during the research, such as academic articles, books, and websites

Allen, J., Charles, B., Fok, C. C. T., Lee, K. S., Grogan-Kaylor, A., & Rasmus, S. (2023). Culturally grounded strategies for suicide and alcohol risk prevention delivered by rural Alaska Native communities: A dynamic wait-listed design evaluation of the Qungasvik intervention. American Journal of Community Psychology, 71(1-2), 184–197. https://doi.org/10.1002/ajcp.12621

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